

MATERIALSVARIANTSSHOE — PARTS ANATOMY

Total Parts: 11



LATERAL SIDE (OUTBOARD)



MEDIAL SIDE (INBOARD)

- ① **Midsole**
Foam layer that provides cushioning.
- ② **Outsole**
Bottom layer that grips the ground.
- ③ **Quarter Panel (Medial)**
Inboard side panel of the upper.

- ④ **Heel Counter**
Stiff support that holds the heel in place.
- ⑤ **Quarter Panel (Lateral)**
Outboard side panel of the upper.
- ⑥ **Toe Box (Vamp)**
Front upper that covers the toes.

- ⑦ **Laces**
Lacing that secures the upper.
- ⑧ **Heel Collar (Topline)**
Padded area around the ankle.
- ⑨ **N Logo (Brand Mark)**
The New Balance "N" on the side.

- ⑩ **Eyelets**
Holes for the laces.
- ⑪ **Tongue Label**
Branding / size label on the tongue.

Dimensions

All dimensions extracted from the 3D mesh after axis canonicalisation and scale calibration. ± values are 2σ bootstrap-resample ranges.

Dimension	Value	Notes
Overall length	270.0 ± 0.2 mm	Toe-to-heel bounding box. ± = 2σ bootstrap range.
Overall width	105.0 ± 0.3 mm	Maximum lateral-medial bbox
Overall height	138.8 ± 0.1 mm	Sole to collar
Outsole thickness	16.7 mm	Top of outsole component above ground
Midsole thickness	31.9 mm	Between outsole top and upper start
Heel stack height	47.4 mm	Sole+midsole height at 20% of length (under heel)
Forefoot stack height	49.7 mm	Sole+midsole height at 80% of length (under metatarsals)
Drop (heel – forefoot)	-2.3 mm	Headline runner-shoe metric. ~10 mm typical for trainers.
Heel height	45.8 ± 2.5 mm	Sole top at the rear 30%
Toe spring	15.0 mm	Lowest point of frontmost 5% of upper above ground
Sole flex point	41.7 %	X-position (% of length) where sole curvature is thinnest
Insole length	244.3 mm	Footbed length at the level of the midsole top
Forefoot girth	230.5 ± 25.8 mm	Alpha-shape perimeter of YZ section at x = +0.65 L
Instep girth	483.0 ± 27.3 mm	Alpha-shape perimeter of YZ section at midfoot (concave-aware)
Throat opening	6.2 mm	Lacing throat width — clearance between lateral & medial uppers
Ankle opening (L × W)	12.5 × 10.2 mm	Collar opening dimensions at the topmost slice
Size: CM (shoe length)	27.0 cm	Last length / 10 — matches NB/Adidas chart 'CM' column
Size: US men's	9.0	1 US = 10 mm of last length
Size: UK men's	8.5	US men's – 0.5
Size: EU	42.5	US men's + 33.5
Foot length (est.)	258.0 mm	Last length – 12 mm toe room

Components (Bill of Materials)

Canonical-vocabulary component list. Confidence ≥ 0.5 means the cluster is a single, real component (high face count, low colour variance, sensible area). Rows below 0.5 are flagged for human review. Perim = cutting-line length used directly for pattern grading.

#	Component	Material	Area (mm²)	Perim (mm)	Color	Conf	Notes
1	eva-midsole	EVA foam midsole	2,564,871 (41%)	7,143	#89A1A7	0.60	—
2	rubber-outsole	rubber outsole	883,720 (14%)	1,386	#F2F2F2	1.00	—
3	quarter-medial	synthetic leather quarter panel	821,295 (13%)	3,076	#5D8691	0.63	—
4	heel-counter	synthetic leather heel counter	629,785 (10%)	1,998	#568593	0.68	—
5	quarter-lateral	synthetic leather quarter panel	619,950 (10%)	2,995	#5E8692	0.62	—
6	vamp	synthetic leather toe-cap	354,179 (6%)	1,643	#5997AA	0.79	—
7	shoe-laces	woven polyester laces	344,464 (6%)	2,331	#6A93A0	0.77	—
8	collar	foam-padded textile collar	42,067 (1%)	524	#79BAD1	0.97	—